## COSMOLOGY A look into the youth of the Milky Way

Galaxies grow by merging with other islands of stars. Our home galaxy was no exception when it formed around 12 to 13 billion years ago. Now Khyati Malhan and Hans-Walter Rix from the Max Planck Institute for Astronomy in Heidelberg have managed to track down two stellar populations that were involved in the earliest of these merger events. They gave the names Shiva and Shakti to the groups, which total around six million stars.

The two researchers evaluated measurement data from the European astrometry satellite Gaia and Him Sloan Digital Sky Survey (SDSS) survey project. At first glance, it may seem surprising that traces of such distant galaxy mergers can still be found. The stars involved, as well as gas and dust accumulations, have long since mixed together. However, the Shiva and Shakti stars have retained some basic properties that go back to the movements of the original galaxies: They each have characteristic values of their angular momentum and kinetic energies that have persisted to this day and that distinguish them from other suns in our galaxy. These values can be used to identify them in the huge data set from the Gaia space telescope, which contains around 1.5 billion entries.

Their chemical fingerprint provides another clue – particularly the content of elements heavier than hydrogen and helium. Malhan and Rix therefore compared the candidates selected from the Gaia data with spectroscopic SDSS data to determine the chemical composition of the celestial bodies. Shiva and Shakti stars are very old and are characterized by low metallicity, i.e. a low proportion of heavy elements.

Two larger groups with a total of six million members stood out, each with similar values for metallicity, kinetic energy and angular momentum. A long time ago they apparently belonged to separate galaxies that have now merged into the Milky Way. Their names have a religious background: Shiva is one of the main deities of Hinduism; Shakti a female cosmic force often depicted as the consort of Shiva.

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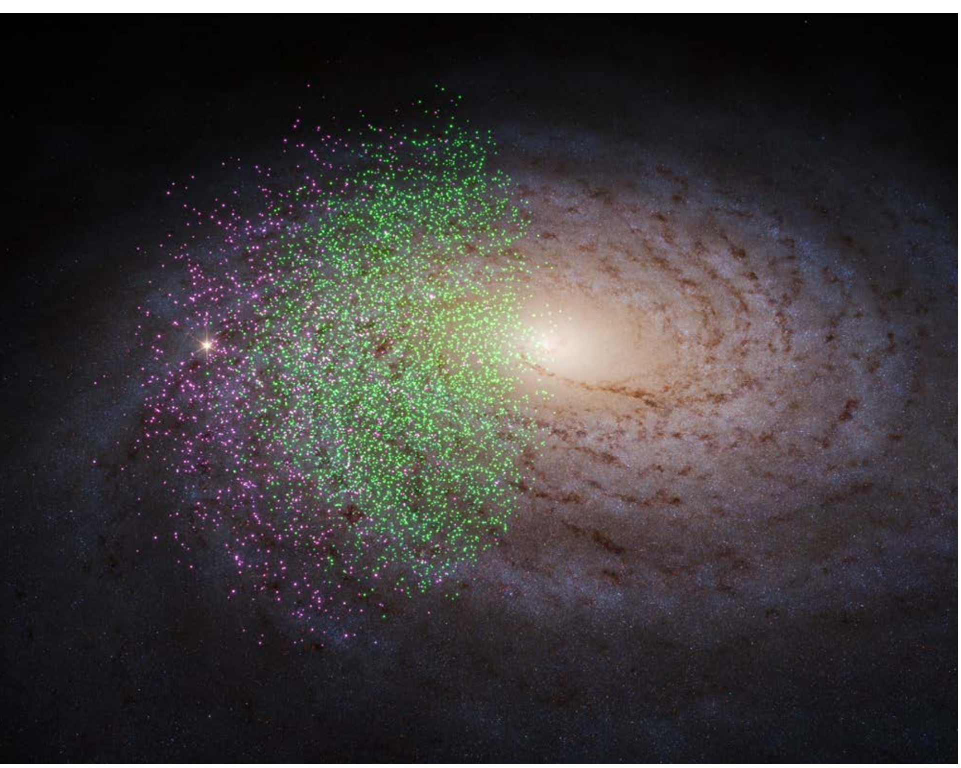

SHAKTI AND SHIVA Based on data from the astrometry satellite Gaia, two ancient star populations can be identified, called Shiva (green) and Shakti (purple). They come from two galaxies that merged with the Milky Way system more than twelve billion years ago.

## NEUROPHYSIOLOGY Our brains are getting bigger

Human brains have grown over the last few generations. Anyone born in the 1960s has a thinking organ that is almost seven percent larger than someone born in the 1930s. This may offer some protection against age-related brain diseases such as Alzheimer's. This is the conclusion reached by a research group led by Charles DeCarli from the University of California in Davis.

The team relied on data from the Framingham Heart Study, a long-term study in the USA. It includes thousands of people from multiple generations and documents